

B. Tech. (4th Sem) (Common for All Streams except CSE-Industry Integrated)

BCSE-505 (Discrete Structures)

L	T	P
3	0	0

Continuous evaluation	40
End semester exam	60
Total marks	100
Credits	3.0

Course Objectives:

1. To get familiar and understand the fundamental notions in discrete mathematics
2. To understand and demonstrate the basic concept of an algorithm and its application in combinatorial mathematics
3. To identify the basic properties of graphs and trees and model simple applications

Unit – I

Set Theory And Logic - Sets, Functions, Relations, Equivalence Relation, Poset. Functions Logic: Propositional Logic, Truth Tables, Tautologies, Resolution Proof System, Predicate Logic.

Unit – II

Induction And Combinatorics - Peano's Axioms - Mathematical Induction - Pigeon-Hole Principle - Principle Of Inclusion And Exclusion - Review Of Permutations And Combinations - Distribution Problems - Derangements - Bijection Principle.

Unit – III

Linear Algebra and Recurrence Relations- Linear Algebra: Vector Space, Basis, Dimension, Orthogonality. Recurrence Relations: Homogeneous and Inhomogeneous Recurrences and Their Solutions - Solving Recurrences Using Generating Functions.

Unit – IV

Graph Theory- Definitions And Basic Results - Representation Of A Graph By A Matrix And Adjacency List - Trees - Cycles - Properties - Paths And Connectedness - Subgraphs - Graph Isomorphism - Operations On Graphs - Vertex And Edge Cuts - Vertex And Edge Connectivity.

Course Outcomes: After completion of this course, student will be able to:

- i) Ability to distinguish between the notion of discrete and continuous mathematical structures
- ii) Ability to construct and interpret finite state diagrams and DFSA
- iii) Application of induction and other proof techniques towards problem solving

Instructions for paper setter: All Questions are compulsory. The Question paper is divided in to four sections A, B, C and D. Section A is compulsory and comprises of 12 questions of one mark each, 3 from each unit. The questions shall be asked in such a manner that there are no direct answers including one word answer, fill in the blanks or multiple choice questions. Section B comprises of 4 questions of 2 marks each, one from each unit. Section C Comprises of 4 questions of 4 marks each, one from each unit. Section D Comprises of 4 questions of 6 marks each, one from each unit. There is no overall choice, however internal choice may be provided in section C and D, if paper setter so desires.

Text Books:

1. C.L.Liu And D.P.Mohapatra, " Elements Of Discrete Mathematics: A Computer Oriented Approach", Mcgraw Hill, Third Edition, 2012.
2. Kenneth H. Rosen, "Discrete Mathematics And Its Applications" Mcgraw Hill, Seventh Edition, 2012 (Indian Adaptation by Kamala Krithivasan, IIT Madras).

Reference Books:

1. R. Balakrishnan and K. Ranganathan, "A Text Book Of Graph Theory", Springer
2. Thomas Koshy, "Discrete Mathematics with Applications", Elsevier, 2009.